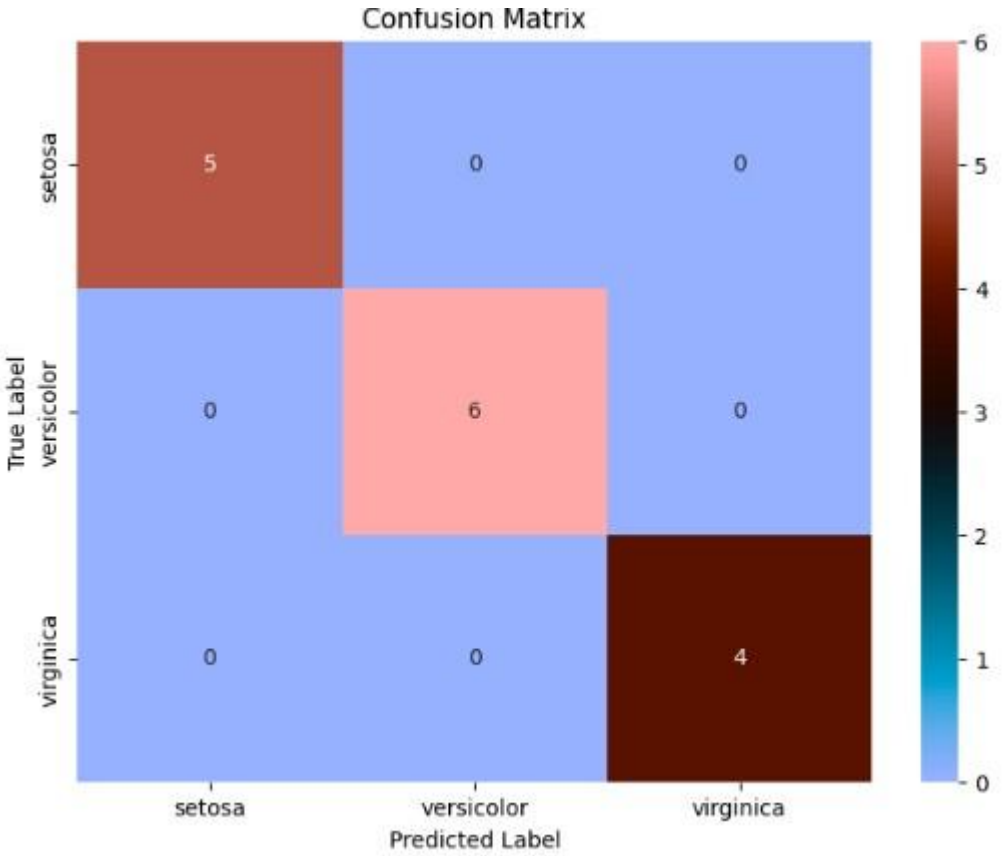
Experiment 11:

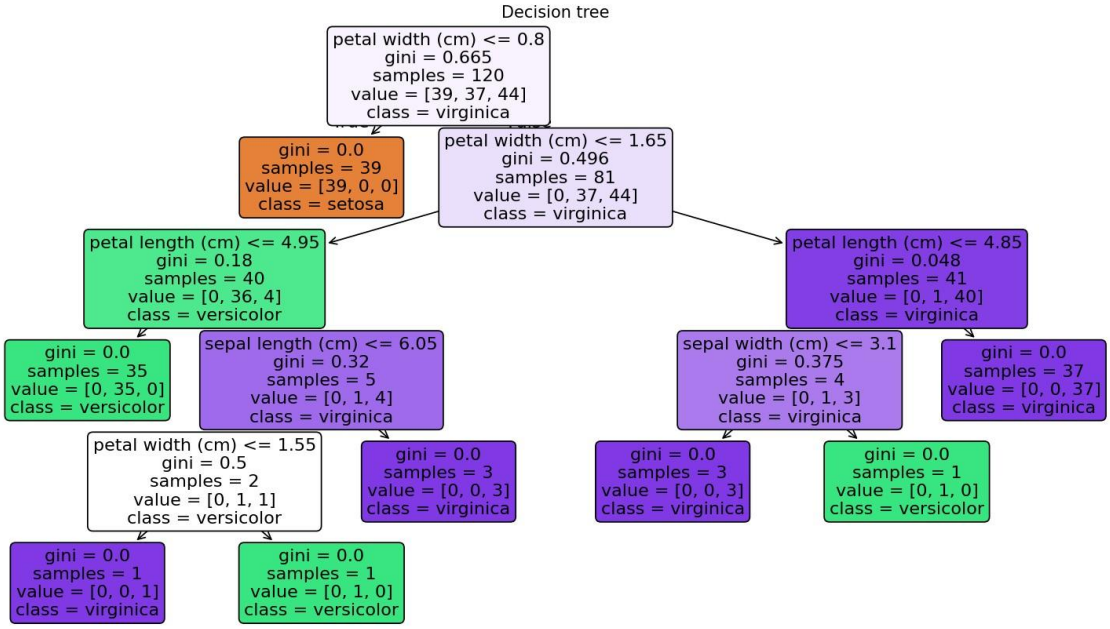
Program:

```
import pandas as pd import matplotlib.pyplot as plt
import seaborn as sns from sklearn.datasets import
load_iris from sklearn.model_selection import
train_test_split from sklearn.tree import
DecisionTreeClassifier from sklearn.metrics import
accuracy_score,confusion_matrix iris = load_iris()

data = pd.DataFrame(data=iris.data,
columns=iris.feature_names) data['species'] = iris.target X
= data.drop('species', axis=1) y = data['species']
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=1) model
=DecisionTreeClassifier() model.fit(X_train, y_train) y_pred
= model.predict(X_test) accuracy = accuracy_score(y_test,
y_pred) print("accuracy:",accuracy) conf_matrix =
confusion_matrix(y_test,y_pred) plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d',
cmap='PRGn', xticklabels=iris.target_names,
yticklabels=iris.target_names) plt.xlabel('Predicted
Label') plt.ylabel('True Label') plt.title('Confusion
Matrix') plt.show()
```

Output:

```
accuracy:
1.0
```



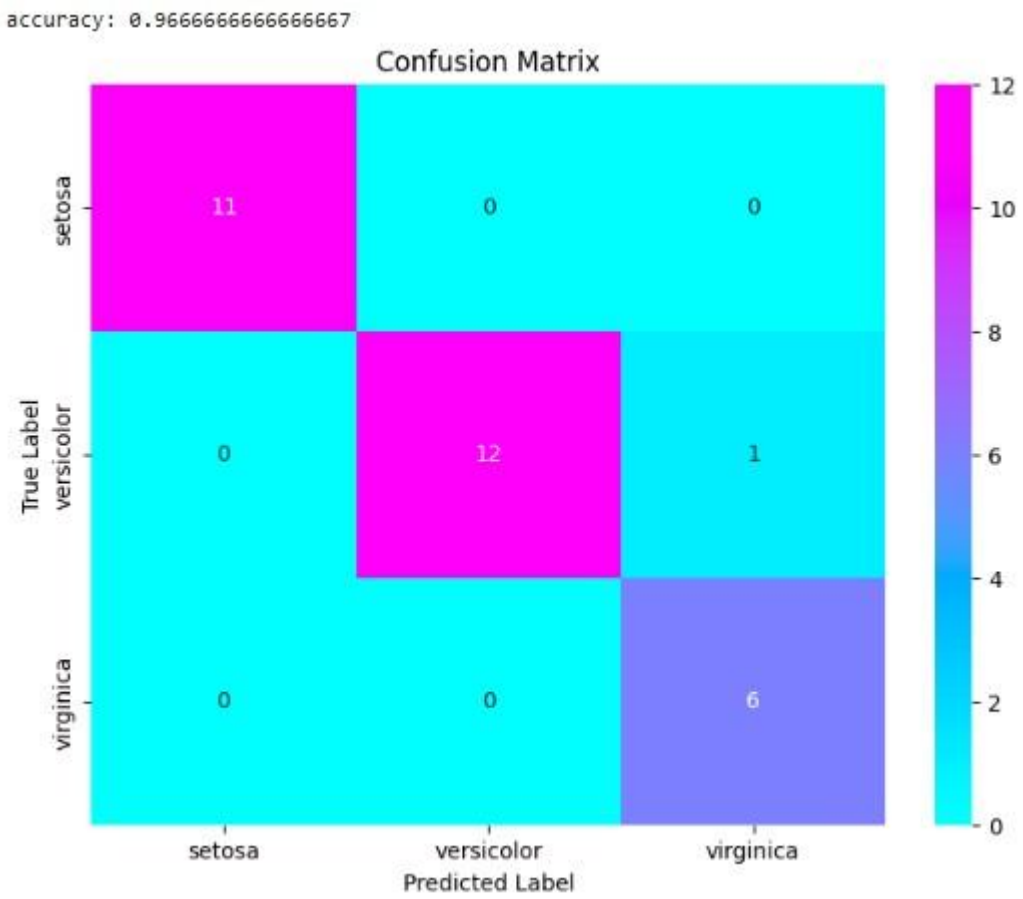


Experiment 12:

Program:

```
import pandas as pd import matplotlib.pyplot as plt import seaborn as
sns from sklearn.datasets import load_iris from sklearn.model_selection
import train_test_split from sklearn.ensemble import
RandomForestClassifier from sklearn.metrics import
accuracy_score,confusion_matrix iris=load_iris()
data=pd.DataFrame(data=iris.data,columns=iris.feature_names)
data['Species']=iris.target x=data.drop('Species',axis=1) y=data['Species']
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_st
ate=1)
model=RandomForestClassifier(n_estimators=100,random_state=1)
model.fit(x_train,y_train) y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred) print("accuracy:",accuracy)
conf_matrix = confusion_matrix(y_test,y_pred) plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='managua',
xticklabels=iris.target_names, yticklabels=iris.target_names)
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.title('Confusion Matrix')
plt.show()
```

Output:



Experiment 13:

Program:

import pandas as pd import matplotlib.pyplot

as plt import seaborn as sns from

sklearn.datasets import load_iris from

sklearn.model_selection import

train_test_split from sklearn.svm import SVC

from sklearn.metrics import

accuracy_score,confusion_matrix

iris=load_iris()

data=pd.DataFrame(data=iris.data,columns=ir

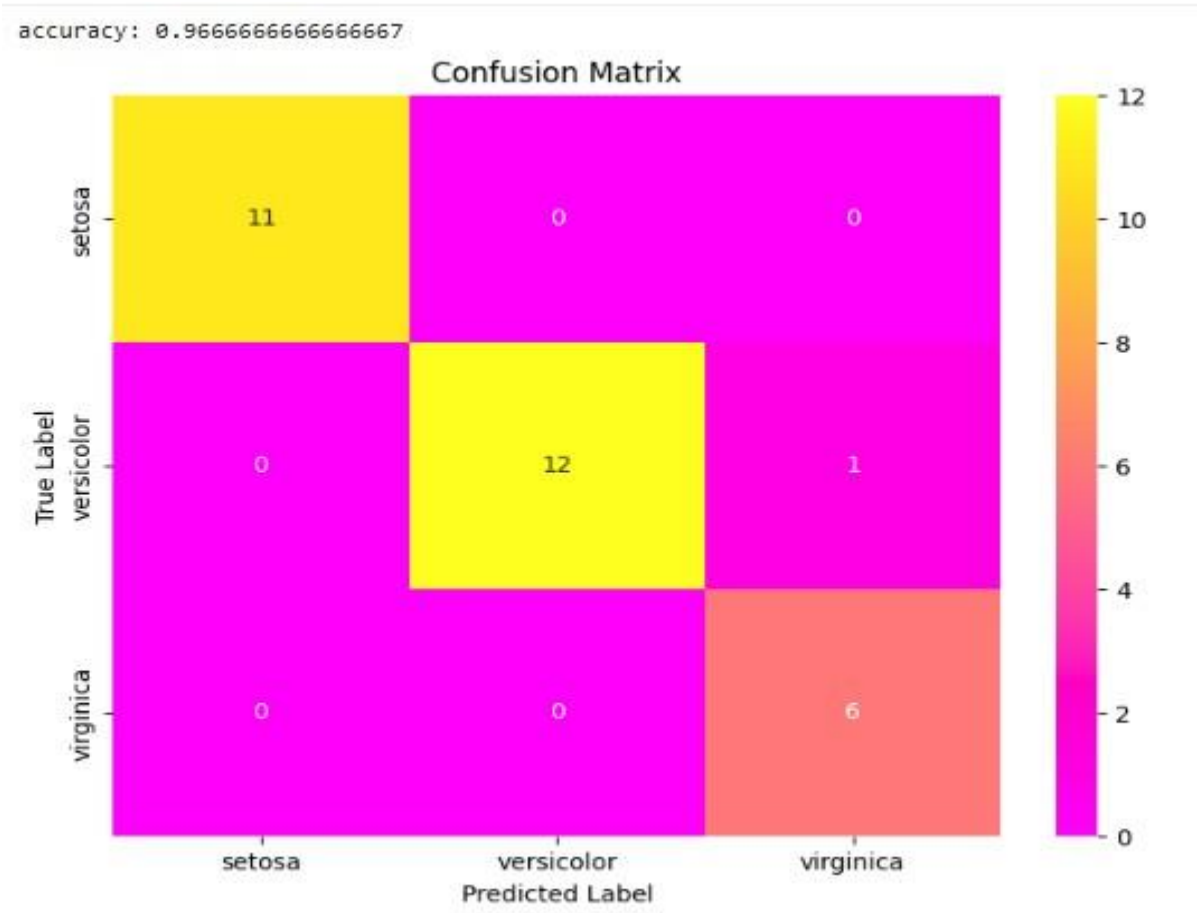
is.feature_names) data['Species']=iris.target

x=data.drop('Species',axis=1)

```
y=data['Species']
x_train,x_test,y_train,y_test=train_test_split(
x,y,test_size=0.2,random_state=1)
model=SVC(kernel='poly',random_state=1)
model.fit(x_train,y_train)
y_pred=model.predict(x_test)
accuracy=accuracy_score(y_test,y_pred)
print("accuracy:",accuracy) conf_matrix =
confusion_matrix(y_test,y_pred)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt='d',
cmap='turbo', xticklabels=iris.target_names,
yticklabels=iris.target_names) plt.xlabel('Predicted
Label') plt.ylabel('True Label') plt.title('Confusion
Matrix') plt.show()
```

Output:

accuracy: 0.9666666666666667



Experiment 14:

Program:

import numpy as np import

matplotlib.pyplot as plt from

sklearn.preprocessing import

StandardScaler def

mean_squared_error(y_true, y_predicted):

    cost = np.sum((y_true - y_predicted)**2) / len(y_true)

return cost

def gradient_descent(x, y, iterations=1000,

learning_rate=0.01, stopping_threshold=1e-6):

current_weight = 0.0   current_bias = 0.0

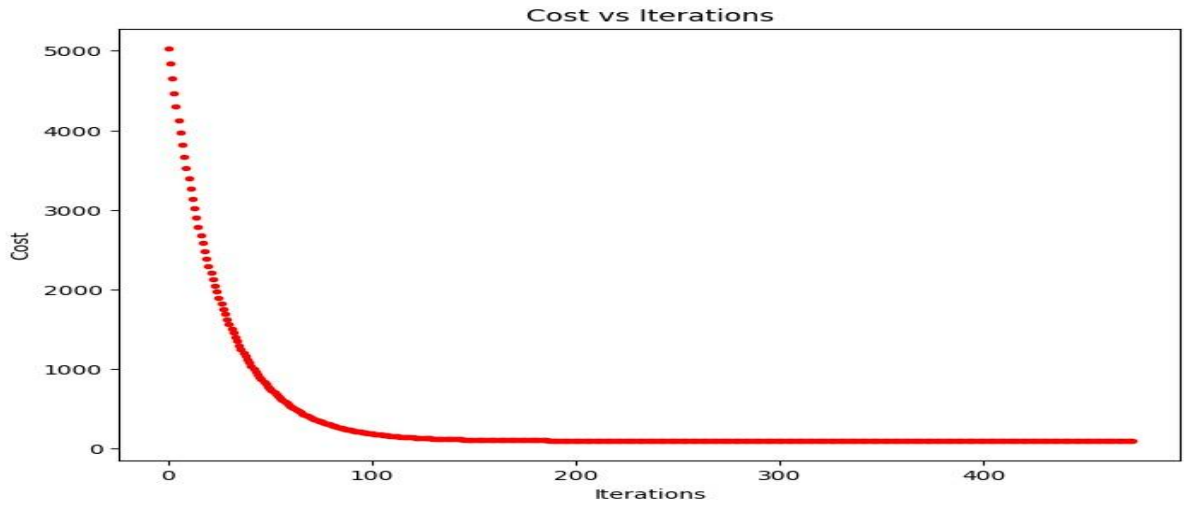
```
    n = float(len(x))
costs = []
previous_cost =
None     for i in
range(iterations):
        y_predicted = current_weight * x + current_bias
current_cost = mean_squared_error(y, y_predicted)
        if previous_cost and abs(previous_cost - current_cost) <=
stopping_threshold:
            break         previous_cost = current_cost
costs.append(current_cost)         weight_derivative = -(2/n) *
sum(x * (y - y_predicted))         bias_derivative = -(2/n) * sum(y -
y_predicted)         current_weight = current_weight - learning_rate
* weight_derivative         current_bias = current_bias -
learning_rate * bias_derivative         if i % 100 == 0:
            print(f"Iteration {i+1}: Cost {current_cost}, Weight
{current_weight}, Bias {current_bias}")     plt.figure(figsize=(8,6))
plt.plot(range(len(costs)), costs, 'r.')     plt.title("Cost vs Iterations")
plt.xlabel("Iterations")     plt.ylabel("Cost")     plt.show()     return
current_weight, current_bias def main():
X = np.array([32.5, 53.4, 61.5, 47.4, 59.8, 55.1, 52.2, 39.2, 48.1, 52.5,
    45.4, 54.3, 44.1, 58.1, 56.7, 48.9, 44.6, 60.2, 45.6, 38.8])
Y = np.array([31.7, 68.7, 62.5, 71.5, 87.2, 78.2, 79.6, 59.1, 75.3, 71.3,
    55.1,
82.4, 62.0, 75.3, 81.4, 60.7, 82.8, 97.3, 48.8, 56.8])
scaler = StandardScaler()
    X_normalized = scaler.fit_transform(X.reshape(-1, 1)).flatten()
```

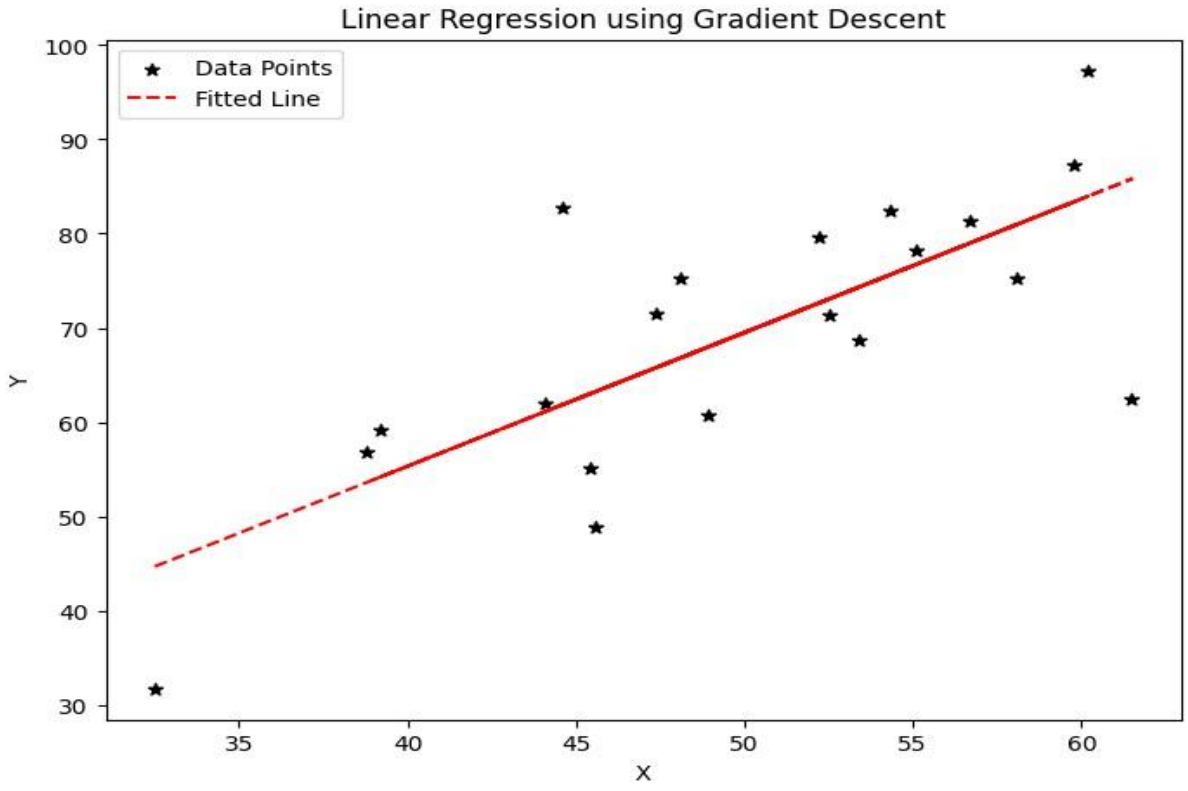
```
    estimated_weight, estimated_bias = gradient_descent(X_normalized, Y,
iterations=2000, learning_rate=0.01)
    print(f"Estimated Weight: {estimated_weight}, Estimated Bias:
{estimated_bias}")
    Y_pred = estimated_weight * X_normalized +
estimated_bias     plt.figure(figsize=(8,6))     plt.scatter(X, Y,
color='black', label='Data Points',marker='*')     plt.plot(X,
Y_pred, color='red', linestyle='--', label='Fitted Line')
plt.xlabel("X")     plt.ylabel("Y")     plt.title("Linear
Regression using Gradient Descent")     plt.legend()
plt.show()

if __name__ == "__main__":
    main()
```

Output:

```
Iteration 1: Cost 5031.3015, Weight 0.21744528996901658, Bias 1.3877
Iteration 101: Cost 185.56941123866557, Weight 9.459227106883091,
Bias
60.3672282719577
Iteration 201: Cost 100.34293399589959, Weight 10.684868107118445,
Bias
68.18906711826675
Iteration 301: Cost 98.84397526476002, Weight 10.847412072256065,
Bias
69.22639591234459
Iteration 401: Cost 98.81761165863254, Weight 10.868968580725983,
Bias
69.36396599633204
```

Estimated Weight: 10.87151030998278, Estimated Bias: 69.38018689350649





Experiment 15:

Program:

import cv2 import numpy as
np from matplotlib import

```
pyplot as plt from
google.colab import drive

# Mount Google Drive
drive.mount('/content/drive')
# For Colab: !wget https://example.com/cataract_image.jpg -O image.jpg
(replace with actual URL) img =
cv2.imread('/content/drive/MyDrive/dog.jpeg')  # Replace with your path

# Check if the image was loaded
successfully if img is None:
   print("Error: Could not load image. Please check the file
path.") else:
   rgb_img = cv2.cvtColor(img,
cv2.COLOR_BGR2RGB)     pixels =
np.float32(rgb_img.reshape((-1, 3)))     criteria =
(cv2.TERM_CRITERIA_EPS +
cv2.TERM_CRITERIA_MAX_ITER, 100, 0.2)
   K = 3
   _, labels, centers = cv2.kmeans(pixels, K, None, criteria, 10,
cv2.KMEANS_RANDOM_CENTERS)
   centers = np.uint8(centers)     segmented_img =
centers[labels.flatten()].reshape(rgb_img.shape)
plt.figure(figsize=(10, 5))     plt.subplot(121)
plt.imshow(rgb_img)     plt.title('Original Image')
plt.axis('off')

   plt.subplot(122)
plt.imshow(segmented_img)
plt.title('Segmented Image (K-
means)')     plt.axis('off')
plt.tight_layout()     plt.show()
```

Output:

Original Image

Segmented Image (K-means)